

Criterion B: Design

Preface

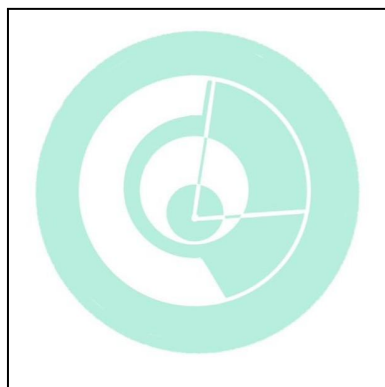
The to-do application with the name: “TickOffTime”.

The programme consists of:

- Creating a task
 - Entering a task’s name
 - Choosing a category a task will be assigned to
 - Choosing the type of a task
 - Adjusting a starting day
 - Adjusting a final day
 - Adjusting a repetition
 - Entering a task’s description
- Editing a task
- Deleting a task
- Creating a category
 - Entering a category’s name
 - Choosing a category’s colour
- Editing a category
- Deleting a category
- Searching and filtering tasks
 - Seeking by a name
 - Choosing a category filter
 - Choosing the type of task filter
 - Choosing a date filter
- Presenting a grouped list of the created tasks
- Presenting a calendar with the assigned to specific day tasks
- Presenting the “To do today” menu
- Presenting the “To do this week” menu
- 2 theme options

The app’s logo

Figure 1. The graphic presenting the design of the logo of the app



Creating a task

Figure 2. The graphic presenting the design of the “creating a task” window

- Only the “Name of a task”, “Description”, “+”, “SAVE” and “<--” buttons are clickable.

- The “Name of task” field allows a user to enter the name of a task.

- The “Description” button enables a user to enter a text that will be attached to the task

- The arrow in the bottom left-hand corner allows a user to return to the menu without saving (Figure 16.).

- The “SAVE” button at the bottom allows a user to go back to the menu and save a task only if the name and the type were set by a user.

- All tasks are stored in a database with specific features. (Table 1.)

Table 1. The description of features of Task

Name of a feature	Description	Type of data	How is it declared?	Initial value
Id	A task’s unique id	Int	Automatically	0
Name	A task’s name	String	By a user	“”
Category	A task’s affiliation for a category	String	From the data set	“Uncategorised”
Type	A task’s type	String	From the data set	“”
Description	A task’s description	String	By a user	“”
First_date	A task’s start day	String	From the data set	Present day (as string)
Second_date	A task’s final day	String	From the data set	999999 days from the present day (as string)
Days_repeated	A task’s days when it will be repeated	Int	From the data set	1234567

Figure 3. The graphic presenting a design in case of clicking the “Category” “+” button

Name of task	
+	Category
+	Category 1
+	Category 2
+	Category 3
+	Category 4
+	Category 5
+	Type

- In the case of clicking the “+” button next to “Category”, all already created categories are shown.
- In the case of a lack of categories, the information “No categories” is displayed.
- In order to choose a category, click the “+” next to it.
- At most one category can be chosen.
- If a user creates a task without assigning it to any category, the task is assigned to the built-in “Uncategorised” category automatically and will appear under its name.

Figure 4. The flowchart presenting the “displaying categories” operation

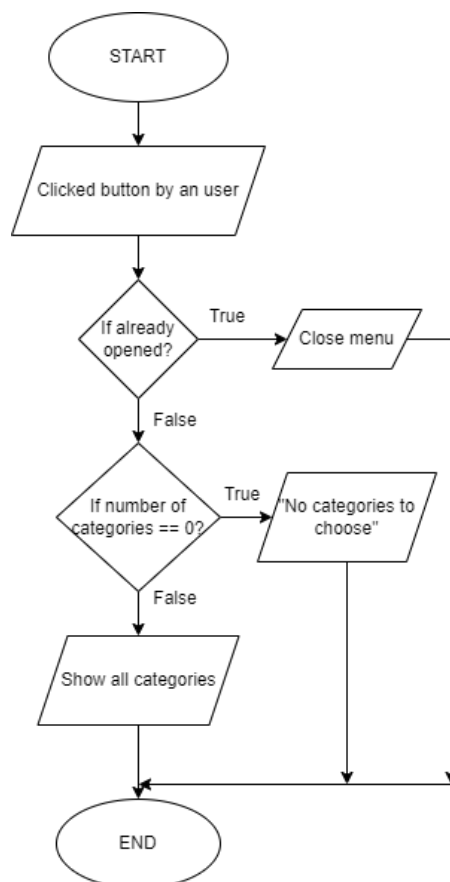


Figure 5. The graphic presenting the design in case of clicking the “Type” “+” button

Name of task							
+	Category						
+	Type						
<table border="1"> <tr> <td>+</td> <td>Single task</td> </tr> <tr> <td>+</td> <td>Until task</td> </tr> <tr> <td>+</td> <td>Repeat task</td> </tr> </table>		+	Single task	+	Until task	+	Repeat task
+	Single task						
+	Until task						
+	Repeat task						
Description							

- In the case of clicking the “+” button next to “Type”, all types are shown.
- In order to choose a type, a user clicks the “+” next to it.
- Exactly one type can be chosen.
- Having chosen the “Until task” or “Repeat task”, 2 new options appear (Figure 6.).

Table 2. The description of types of tasks

Name of a type	Description	Required number of chosen days	First choice/Since	Second choice/To
Single task	A task will be assigned to only one day without repetition	1	The day a task will be assigned to	n/a
Until task	A task will be repeated until the final day (called a deadline) comes	2	Since when a task should be repeated	The deadline
Repeated task	A task will be repeated regularly only on chosen days	2	Since when a task should be repeated	Until when a task should be repeated

Figure 6. The graphic presenting the design of additional options for Until and Repeat tasks

Name of task	
+	Category
+	Type
+	Single task
+	Until task
+	Repeat task
+	Until
+	Repeat every

- Having clicked the “+” next to “Until” or next to “Single task”, a calendar with an opportunity to choose at most two dates appears (Figure 7.). In the case of the Single task, a user can choose only one date.
- Having clicked the “+” next to “Repeat every”, an array with an opportunity to choose days when a task will be repeated regularly appears. (Figure 8.)

Figure 7. The graphic presenting the design of the “choosing a date” window

<	August							>
Mon	Tue	Wed	Thu	Fri	Sat	Sun		
	1	2	3	4	5	6		
7	8	9	10	11	12	13		
14	15	16	17	18	19	20		
21	22	23	24	25	26	27		
28	29	30	31					
Since		To						
←	SAVE							

- The buttons “<” and “>” are clickable and allow a user to proceed to the previous or the next month consecutively.
- Grey numbers signify past days and are not clickable.
- Having clicked the “Since” button a user can choose the start day of a task
- Having clicked the “To” button a user can choose the end day of a task called a deadline.
- The arrow in the bottom left-hand corner allows you to return to the previous window without saving (Figure 6.).
- The “SAVE” button at the bottom allows you to return to the menu and save chosen days (Figure 6.)

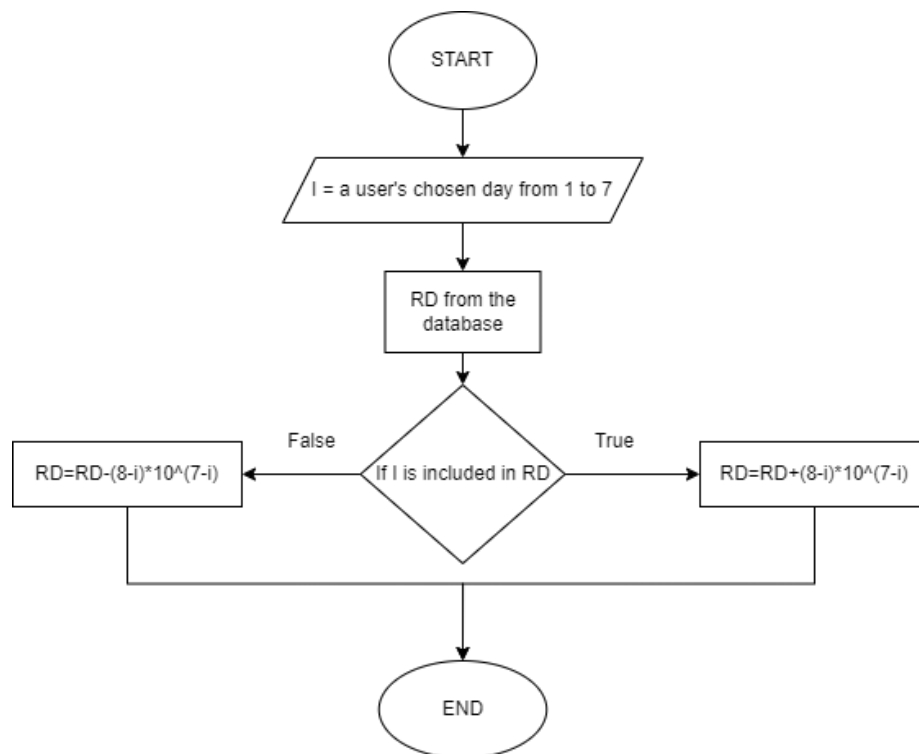
Figure 8. The graphic presenting the design of the “choosing days” option



- At least one day must be chosen.
- The chosen days are in a dark mint colour.
- A value of 1234567 signifies that a task will be repeated every day and each digit signifies a day of the week (Friday =

5). Nonetheless, the days a user wishes to disregard are signified by 8, which does not signify any day but informs a day should be omitted. In this example only Tuesday, Friday, and Sunday are not chosen thus this value is 1834868.

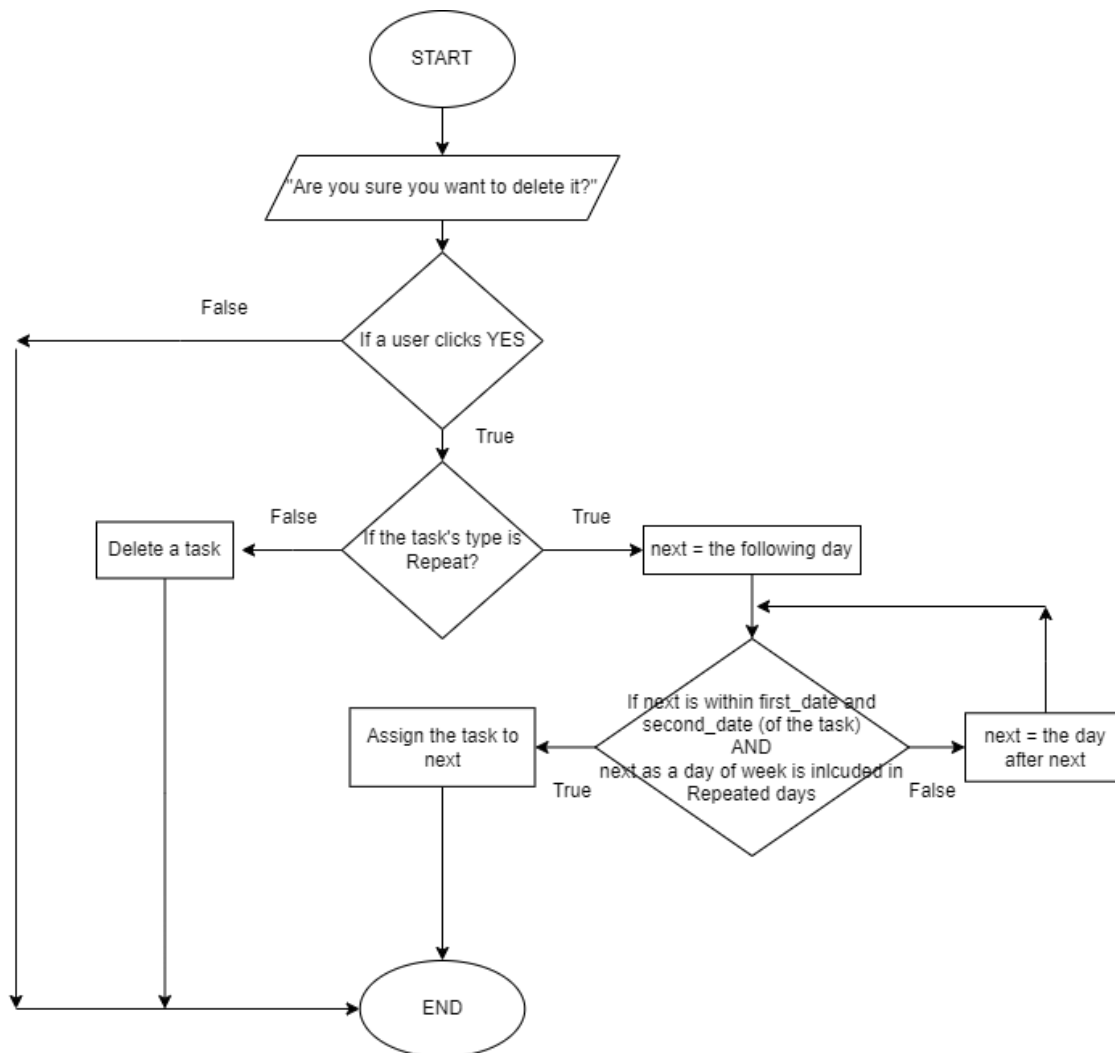
Figure 9. The flowchart presenting the “Repeated days” operation



Done tasks

- When a task is considered to be done (this situation is described later), the Until and Single tasks are deleted. Nonetheless, the Repeat tasks are reassigned to the first possible day after the present day.

Figure 10. The flowchart presenting the “done tasks” operation



Editing a task

Figure 11. The graphic presenting the design of the “editing a task” window

Name of task	
+	Category
+	Type
Description	
Delete	
←	SAVE

- In the case of the “editing” view, the “Name of task” and “Description” fields are already filled based on the features of the task.
 - A user can change each feature and save in the same way as in the “creating” operation.
 - The mint “Delete” button allows a user to delete a task permanently regardless of its type.
 - The arrow in the bottom left-hand corner allows a user to return to the previous window without saving changes.
 - The “SAVE” button at the bottom allows a user to return to the menu and save changes.

Reading a task

Figure 12. The graphic presenting the design of the “reading a task” window

Name of task	
Description	
←	

- In this case, a user can only read a task’s name and description without the possibility of changing it.
 - The arrow in the bottom left-hand corner allows you to return to the previous window without saving changes.

Creating a category

Figure 13. The graphic presenting the design of the “creating a category” window

Hand-drawn UI design for the "creating a category" window. The window contains the following elements:

- A text input field labeled "Name of category".
- A second text input field labeled "Colour of category" with a small "+" icon to its left.
- A list of five colored bars (red, yellow, green, blue, and purple) below the "Colour of category" field.
- A back arrow icon at the bottom left.
- A "SAVE" button at the bottom right.

- In this window, a user is able to create a category.
- A user enters a category's name and chooses its colour.
- The rest works in the same way as the “Creating a task” operation (Figure 2.).
- All categories are stored in a database with specific features. (Table 3.)

Editing a category

Figure 14. The graphic presenting the design of the “editing a category” window

Hand-drawn UI design for the "editing a category" window. The window contains the following elements:

- A text input field labeled "Name of category".
- A second text input field labeled "Colour of category" with a small "+" icon to its left.
- A "Delete" button below the "Colour of category" field.
- A back arrow icon at the bottom left.
- A "SAVE" button at the bottom right.

- In the case of the “editing” view, the “Name of category” field is already filled based.
- A user can change each feature and save in the same way as in the “creating” operation.
- The mint “Delete” button allows a user to delete a category permanently.

Figure 15. The flowchart presenting the “deleting a category” operation

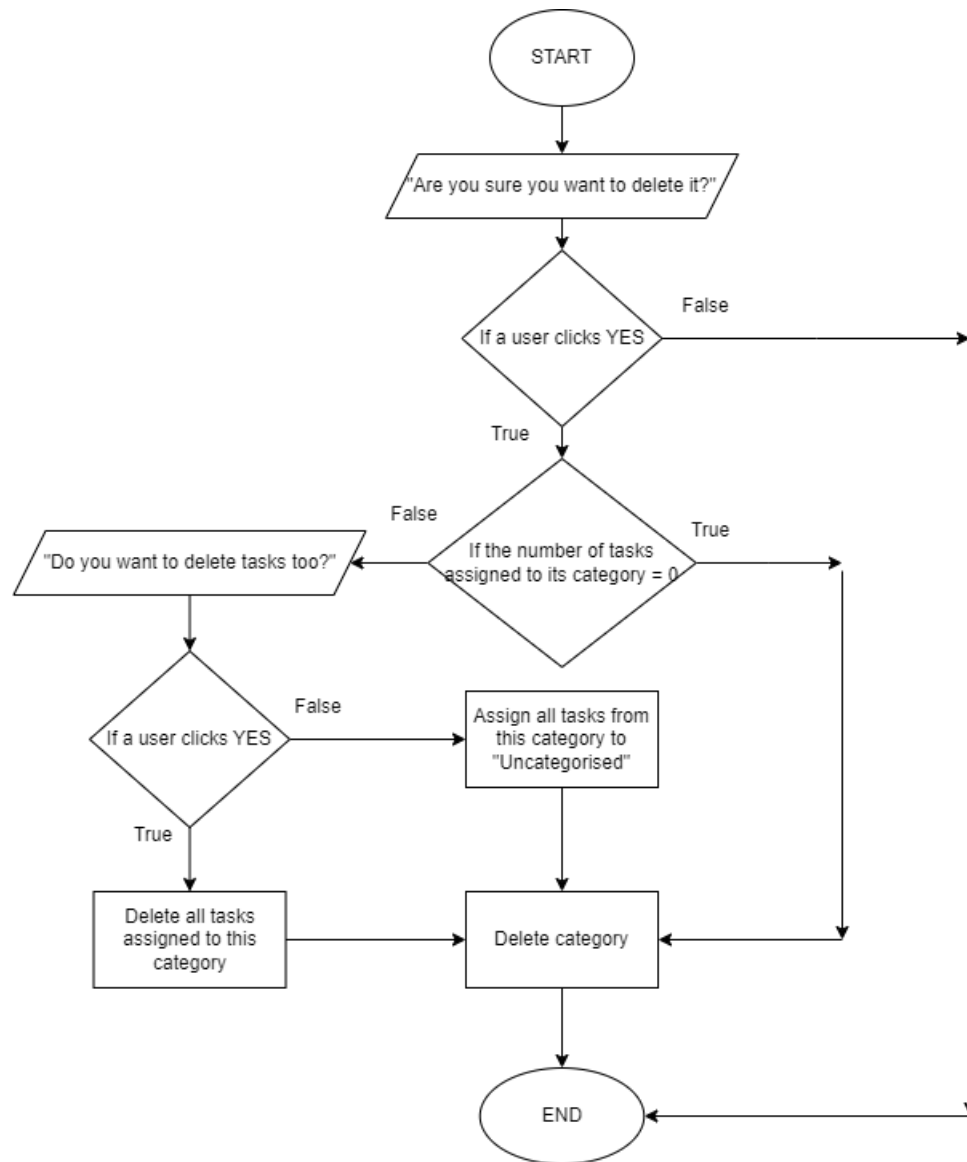


Table 3. The description of features of Category

Name of a feature	Description	Type of data	How is it declared?	Initial value
Id	A category's unique id	Int	Automatically	0
Name	A category's name	String	By a user	""
Colour	A category's colour	Int	From the data set	0

List of tasks

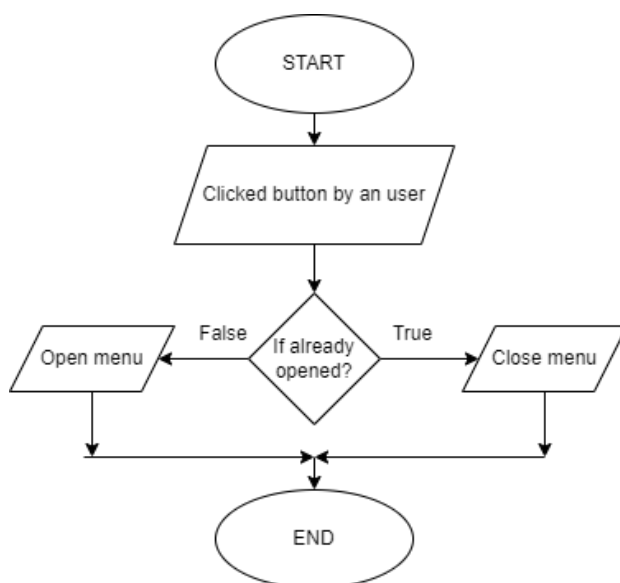
Figure 16. The graphic presenting the design of the “List of tasks”.

- Having clicked on a button in the **bottom left-hand corner**, a list of all created tasks appears in the form of a scrollable column.
- Having clicked the **mint “+” button** on the right-hand side, a new window appears which allows a user to create a task (Figure 2.)
- All tasks are grouped into categories that they have been assigned to.
- Having clicked the **“>” button**, all assigned tasks appear in order based on the day they are assigned to.
- Each type has its unique colour.
 - **Single tasks**
 - **Until tasks**
 - **Repeat tasks**
- Each category has its colour and name chosen by a user.
- Having clicked the **mint “+” button** twice, a new window appears which allows a

user to create a category (Figure 13.).

- Having clicked the check mark, a task is considered to be done.
- Having clicked the **button “Search”** a new window appears. (Figure 18.).

Figure 17. The flowchart presenting the “opening and closing” operation



Searching a task

Figure 18. The graphic presenting the design of the “searching” window

Search	
Name of task	
+	Category
+	Type
+	Date

- Each row corresponds to the specific features of tasks.
 - Having clicked “**Name**”, a user is able to enter the name of the sought task.
 - Having clicked “**Category**”, the list of categories is shown in the same way as in Figure 3.
 - Having clicked “**Type**”, a new list of types is presented.
 - Having clicked “**Date**”, a calendar appears in the same way as in Figure 7.
 - The chosen options are **bolded**.
 - The mint button in the bottom right-hand corner enables you to go back to the List of tasks (Figure 16.)
- Having clicked the “**Search**” button, the chosen options are saved and used to filter tasks that are presented in a new window. (Figure 19.)

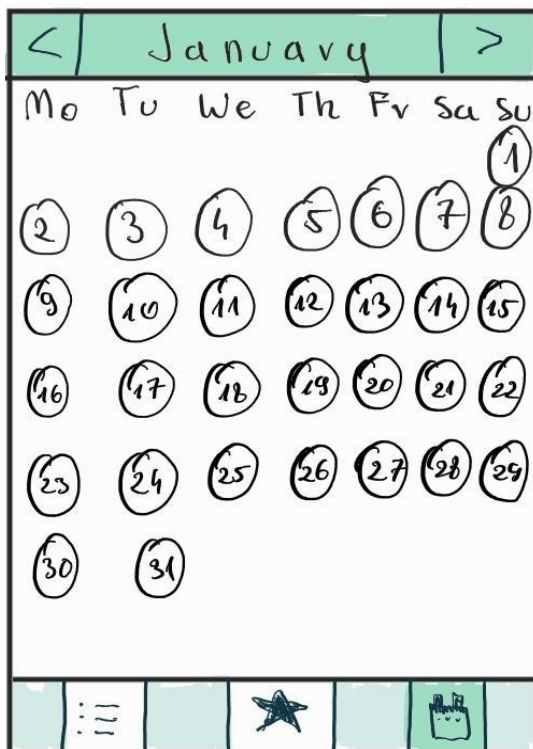
Figure 19. The graphic presenting the design of the “filtered tasks” window

Search	
✓	Task 2
✓	Task 4
✓	Task 5
✓	Task 7

- The “<” button in the bottom right-hand corner enables you to go back to the “filters” window (Figure 18.)
- The tasks are in a colour assigned to their category.
- The tasks appear in chronological order (from the first chosen day to the last one)

Calendar

Figure 20. The graphic presenting the design of a calendar

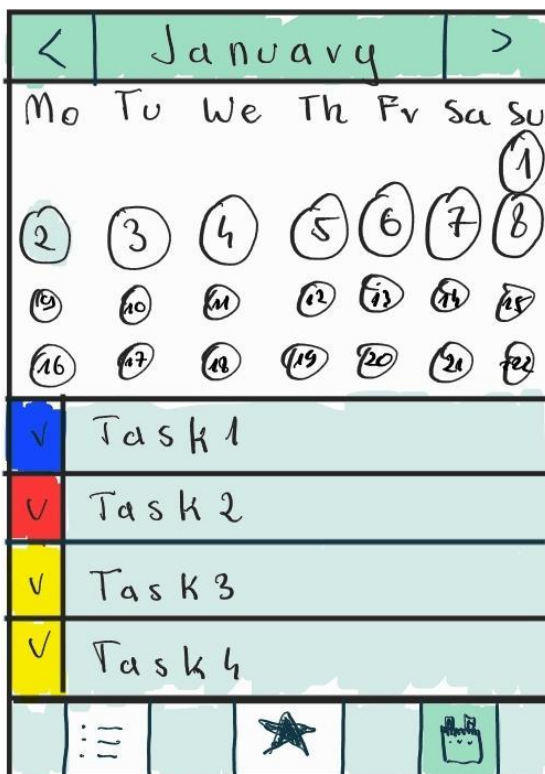


- The buttons in the top left and right corners enable you to proceed to different months.
- A day in mint colour signifies the present day.
- Having clicked a day, all planned tasks on that day appear (Figure 21.).
- All tasks appear in the following order:
 1. **Overdue tasks**
 2. The rest of the tasks
 3. **Until tasks with a coming deadline**

It does not focus mainly on categories, the rest of the tasks can be in random order.

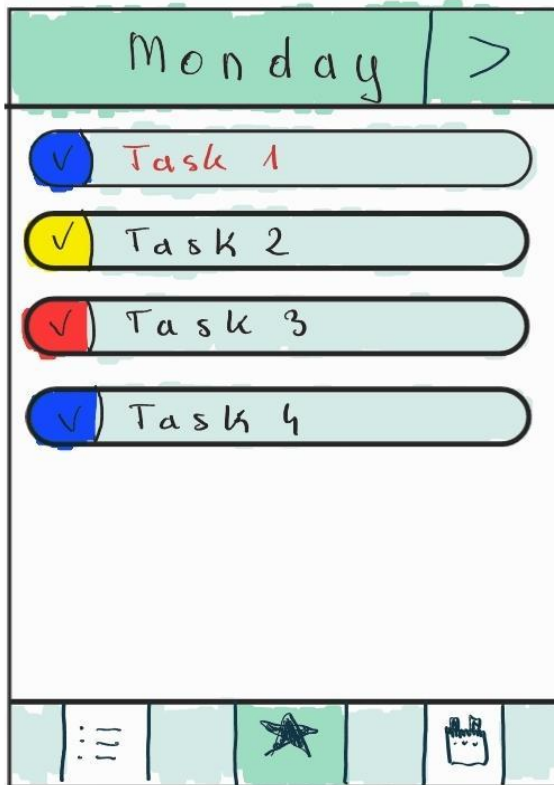
- The tasks are in the colour assigned to their category.

Figure 21. The graphic presenting the design of a calendar after clicking a day



“To do today” menu

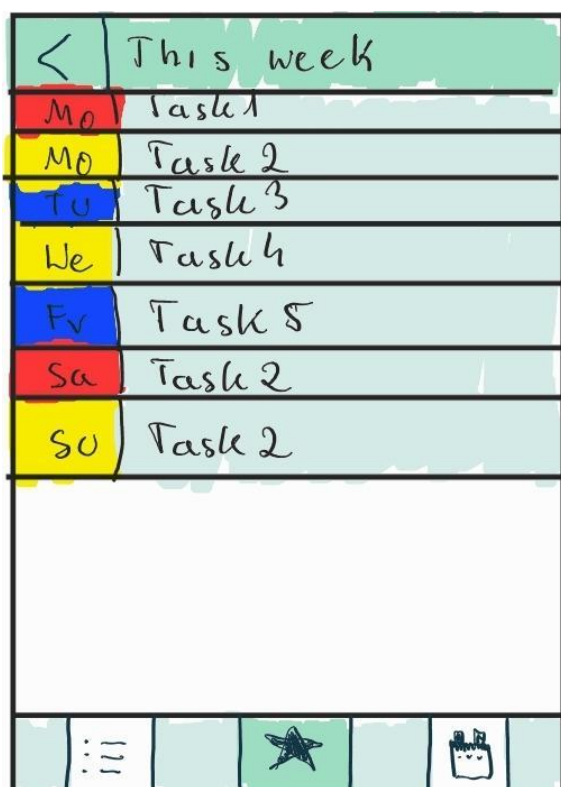
Figure 22. The graphic presenting the design of the “To do today” menu



- This window is considered to be the start destination (it is always open as first)
- There is the present day's name at the top.
- Tasks are arranged in the same order as it was in the calendar.
- Tasks are in the colour assigned to their category.
- Overdue tasks are **red** and until tasks with the coming deadline are **bolded**.
- Having clicked a check mark, a task is considered to be done.
- **The button in the top right corner** moves a user to the “To do this week” menu (Figure 23.).

“To do this week” menu

Figure 23. The graphic presenting the design of the “To do this week” menu



- **The triangle in the top left corner** moves a user to the “To do today” menu. (Figure 22.)
- Tasks are grouped and appear in order based on the day they have been assigned to.

Type scale

Headline	Font	Size	Case
H1	Rosarivo	40sp	The names of months and days
H2	Acme	25sp	The names of the categories
Body1	Tajawal	20sp	The names of the tasks
Body2	Arvo	20sp	Presenting information and announcements, buttons

Colour palette

Slot	Light theme	Dark theme
Primary	#3FDEAF	#202124
Surface	#F8F9FA	#129EAF
Background	#F7F2F2	#007B83
Secondary	#A8F0DA	#F8F9FA
onSurface	#0E1F2F	#F8F9FA
onPrimary	#0E1F2F	#F8F9FA

Test plan

Test type	Nature of test	Example
Test the display of the colour theme	Depending on a theme set by a user, the app has different colours.	A user uses a dark theme so the app starts using specific darker colours. Similarly for a light theme.
Test the display of fonts	The set fonts are displayed correctly depending on the context.	Being in a calendar window you can see “January” in Rosarivo font.
Test the display of tasks	All types of tasks have their proper design: - tasks in the “List of Tasks” menu are in mint colour with the checkmark in one of three colours	Let “Buy flowers” be a single task planned on Tuesday and assigned to a red category “Important”. In the “List of Tasks” it should be in mint colour with the “Buy flowers”

	(purple, light blue, or yellow) + their names. - tasks in the “Search”, “Calendar” and “To Do Today” menus are in mint colour with the checkmark in the category’s colour + their names. - Tasks in the “To Do This Week” menu are in mint colour with the abbreviations for the days of the week in the category’s colour + their names.	name and the checkmark in light blue colour. In the “To Do This Week” it should be in mint colour with the “Buy flowers” name and the abbreviation for Tuesday (“Tue”) in red colour. In other cases, it should be in mint colour with the “Buy flowers” name and the checkmark in red colour.
Test the display of the “Creating a task” menu	The “Creating a task” menu is displayed correctly in appropriate colours. It contains: - the empty “Name of task” field - the mint “Category” bar with the possibility to unroll and display possible colours to choose from. - the mint “Type” bar with the possibility to unroll and display possible colours to choose from. - the mint clickable “<” and “SAVE” buttons at the bottom of the screen.	n/a
Test creating a task	Displayed buttons allow a user to choose features. They are saved properly. If a user does not enter the name of a task or they do not choose the type of the task, they should not be able to save it.	If a user creates a single task “Buy flowers” assigned to the category “Important”, they should be able to save it after clicking the “SAVE” button. If a name or a type is missing, a task should not be saved.
Test the calendar	The calendar is operational, allows a user to proceed to other months, and is clickable. Clicking the chosen day, it becomes mint. After clicking a different day, the previous one becomes white and the new one becomes mint.	Let August be the present month. After clicking the “>” button in the top right corner, the name should change to “September” and all displayed days below should adjust based on the current month. Conversely for the “<” button. Clicking the “2”, it becomes mint. After clicking the “23”, the “2”

		becomes white and the “23” becomes mint.
Test the option allowing the choice of two dates	Having clicked on the button, a calendar appears with two buttons below. A user can choose 2 dates. Chosen dates are saved.	If a user chooses 1.08 as the SINCE and 8.08 as the TO, these features should be saved properly.
Test the option allowing the choice of weekdays	Having clicked on the button, a row appears with 7 clickable abbreviations of weekdays. Chosen days are saved.	If a user chooses only Monday and Sunday, these features should be saved properly as 1888887.
Test editing tasks	Having clicked on the task, the Creating a task menu with already saved features should appear with the possibility to edit and save. They should appear properly with new changes in the “List of Tasks” menu (their name, category, colour, type, or description).	If a user edits the task “Buy flowers” assigned to the category “Important” to the until task “Buy flowers and chocolate”, they should see a new change in the “List of Tasks” menu (a new name and colour).
Test the “Done” option	Having clicked the “Tick” button, a confirmation window appears with 2 possible answers: Yes or No, where the answer “Yes” deletes a task. In case of repeat tasks, they are not deleted but assigned to another possible day.	If a user ticks off the single task “Buy flowers”, a confirmation window should appear. If a user confirms it, the task should be deleted. In another case if the task “Buy flowers” was a repeat task, it should be reassigned to another possible day. As a result, it should not appear in the “To Do Today” menu.
Test deleting tasks	The “Delete” button in the edit view should allow a user to delete a task. The deleted task disappears permanently. Having clicked the “Tick” button, a confirmation window appears with 2 possible answers: Yes or No, where the answer “Yes” deletes a task.	If a user wants to delete the task “Buy flowers”, a confirmation window should appear. The task should disappear permanently only if a user decides to delete it.

Test the display of categories	A programme should display either the scrollable list of categories or the message “No categories”. Categories should be in mint colour with the “>” in colour assigned to this category + their name.	Let “Important” be a category with assigned red colour. It should be displayed at the top of the screen under the “Search” bar and it should be mint colour with the “Important” name and the “>” in red colour. If a user has not created a category yet, they should see the “No categories” message.
Test the display of the “Creating a category” menu	The “Creating a task” menu is displayed correctly in appropriate colours. It contains: - the empty “Name of category” field - the mint “Colour of category” bar with the possibility to unroll and display possible colours to choose. - the mint clickable “<” and “SAVE” buttons at the bottom of the screen	n/a
Test creating a category	Created categories are saved with chosen features and displayed properly in the “List of Tasks”. It is not possible to create a category that has not a name. A user can choose a colour and enter the name.	If a user creates a category “Important” with the chosen red colour, they should see it in the “List of Tasks” menu after clicking the “SAVE” button. If a user chooses a colour without entering a name, they should not be able to save the category.
Test editing categories	Categories open an adequate menu for each. Edits are implemented and saved. The assigned tasks’ category’s name is changed to the category’s new name.	If a user edits the red category “Important” to the yellow category “Important - school”, they should see a new change in the “List of Tasks” menu (a new name and colour). Moreover, all tasks assigned to this category are now assigned to “Important - school” and are displayed in other menus in yellow colour.

Test the category “Uncategorised”	This category does not appear in the List of tasks when it is empty. This category is automatically created while creating a task. This category is not presented as an option when choosing a category as a task feature. The category is uneditable. All tasks without assignment to any existing category are automatically assigned to the “Uncategorised”.	If a user creates a task “Go for a walk” without assignment to any category, they should see it under the “Uncategorised” bar. When a user decides to delete a category but keep tasks, they should see it under the “Uncategorised” bar. If there are no existing tasks or categories, they should see “No categories” in the “List of Tasks” instead of the “Uncategorised”
Test deleting categories	The “Delete” button in the edit view should allow a user to delete a category. The deleted category disappears permanently. If a category is empty, it requires only one confirmation, otherwise, it asks another time if a user wants to delete all assigned tasks as well. If not, they should be reassigned to the “Uncategorised” category.	If a user wants to delete the category “Important” in which there are already tasks, a user should confirm it twice. If a user decides to delete them, they should disappear with a category, otherwise, a user should be able to find them under the “Uncategorised” category. In another case when a category is empty, a user only has to confirm it once.
Test the display of the “List of Tasks” menu	The “List of tasks” is displayed properly in appropriate colours. It contains: - the mint “Search” button on top of the screen - the fundamental main bar at the bottom of the screen with 3 icons, where only the first one is mint - a blank space prepared for categories and tasks	n/a
Test the display of tasks in the “List of Tasks” menu	Tasks should appear only when the categories are expanded and grouped with the category they have been assigned to in the “List of Tasks” menu.	The created task above should appear under the “Important” category bar in light blue colour.
Test the “Search” menu	Searching menu should contain the clickable “Search” button, an empty text field for the name of a	If a user wants to find a task from the category “Important”, which is planned for the following Tuesday. They should be able to choose those

	task, the list of categories, the list of types, a calendar, and a button that allows returning to the List Of Tasks menu. All chosen features should be saved properly.	options, which will be saved properly.
Test the “Showing Results” menu	Only tasks with the chosen features appear. Having clicked the floating tasks, a user should return to the previous menu.	If a user has chosen the features described above, the task “Buy flowers” should appear.
Test the display of the “Calendar” menu	The “Calendar” menu is displayed correctly in the appropriate colours. It contains: - the present month’s name bar with two arrows on top of the screen - the fundamental main bar at the bottom of the screen with 3 icons, where only the third one is mint - a blank space prepared for tasks and a calendar	If the present month is August, a user should see “August” at the top of the screen.
Test the display of tasks in the “Calendar” menu	Having clicked a day in the calendar, a list of tasks based on the chosen day appears.	If a user clicks on the “2” in August, they should see all tasks planned for this day.
Test the display of the “To Do Today” menu	The “To Do Today” menu is displayed correctly in appropriate colours. It contains: - the present weekday’s name on top of the screen - the fundamental main bar at the bottom of the screen with 3 icons, where only the second one is mint - a blank space prepared for tasks	If the present day is Tuesday, a user should see “TUESDAY” at the top of the screen.
Test the display of tasks in the	In the “To Do Today” menu only those tasks should appear that	If a user creates a single task planned for the present day, they should see it in the “To Do Today”

“To Do Today” menu	fulfill one requirement - must be assigned to this day. Additionally, they are grouped by a specific category.	menu.
Test the display of the “To Do This Week” menu	The “To Do This Week” menu is displayed correctly in appropriate colours. It contains: - the “This week” bar on top of the screen - the fundamental main bar at the bottom of the screen with 3 icons, where only the second one is mint - a blank space prepared for tasks	n/a
Test the display of tasks in the “To Do This Week” menu	In the To Do This Week menu only those tasks that fulfill one requirement - must be assigned to the day that is contained in this week should appear. Additionally, they are grouped by the specific day of the week.	If a user plans a single task “Buy flowers” on Tuesday, it should appear only once with the abbreviation “Tue” and if a user has planned a repeat task “Do homework” scheduled for every day, it should appear every day.
Test the functionality of the navigation	Being able to navigate from one menu to another by clicking a proper button.	Seeing a calendar menu, a user can proceed to the list of tasks by clicking its icon.
Test the “Overdue” feature	Overdue tasks’ font is red and they are presented only the present until they are done in all menus except the “List Of Tasks” menu. It is owned by a task whose deadline passed.	If a user has an overdue task “Buy flowers”, it should be in red colour and presented in all menus except the “List of Tasks” menu
Test the “Coming Deadline” feature	Coming Deadline tasks’ font should be bolded and red. It is only presented in the present day. It is visible in all menus except the List Of Tasks menu. It is owned by an until task whose deadline is the next day.	If a user has a coming deadline for the task “Do homework”, it should be in red colour and bolded and presented in all menus except the “List of Tasks” menu.